

PROJECT APPRAISAL

Module 1

Introduction to Project Appraisal



Outline – Module 1

- Cost-benefit analysis frameworks
- Definitions of key concepts
- Cost-benefit analysis model
- Project referent groups
- Project development process
- Project appraisal process: Steps in a project appraisal
- Financial appraisal versus economic/efficiency appraisal
- Project appraisal using spreadsheets: A step-by-step approach
- An illustrative example

Introduction to project appraisal

Cost-benefit analysis frameworks



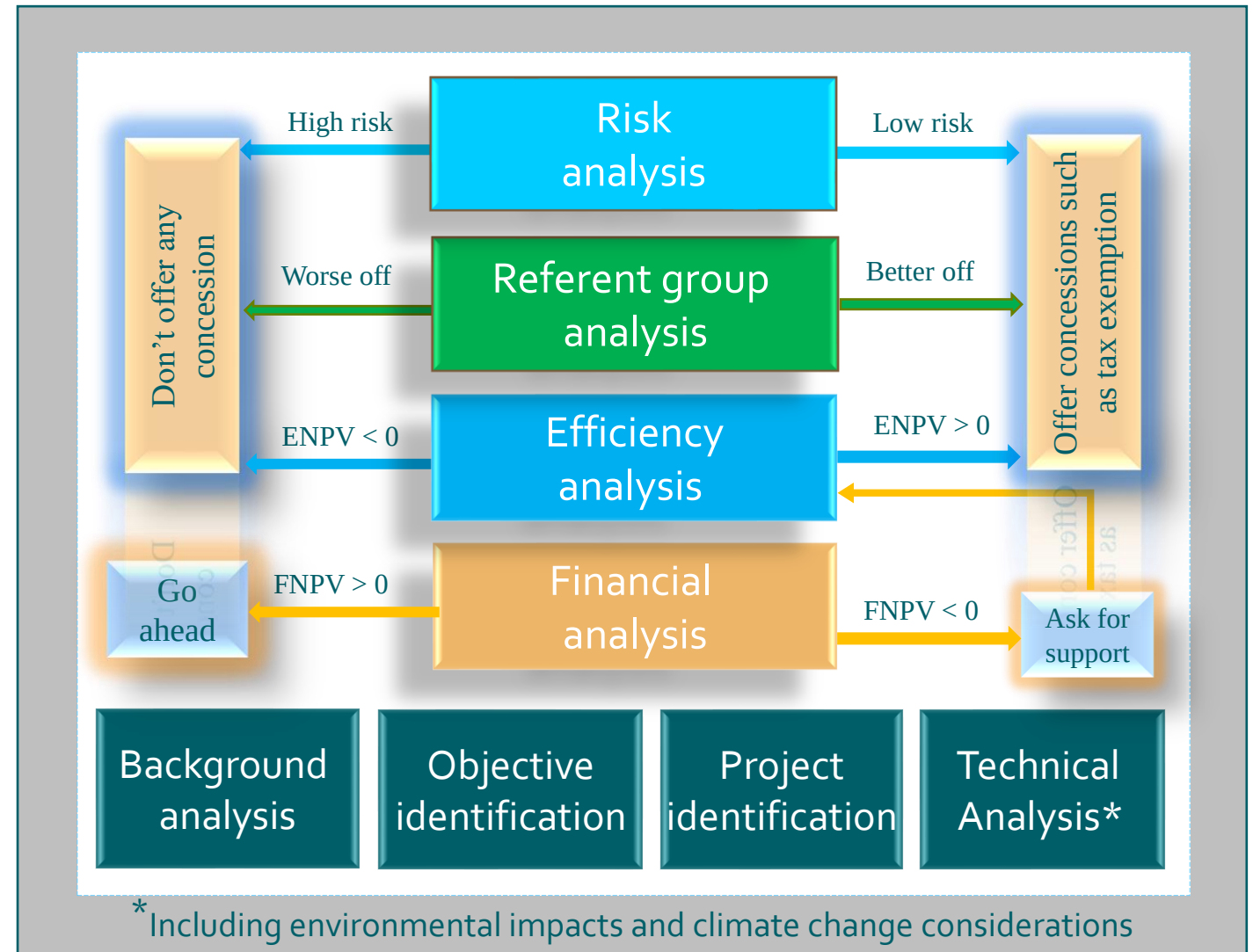
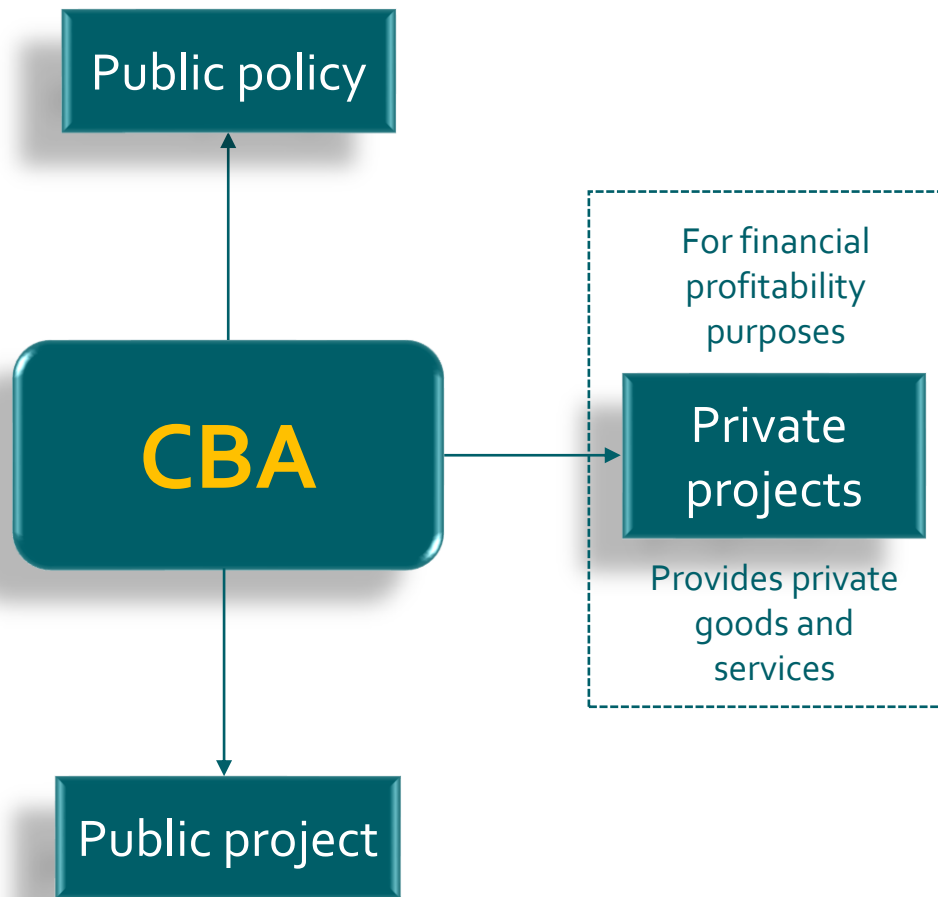
What is cost-benefit analysis?

Cost-benefit analysis (CBA) is a process of identifying, measuring and comparing the benefits and costs of a project (program/policy) undertaken over a period of time from different points of view.

CBA of an **investment project** provides information for decision makers about the level and distribution of benefits and costs, and potentially to contribute to informed public opinion and debate. Ideally, the decision-makers will take the results of the analysis, together with other information, into account in coming to a decision about the project.

The project or projects in question may be **private** or **public** projects; both types of projects need to be appraised to determine whether they represent an efficient use of scarce resources.

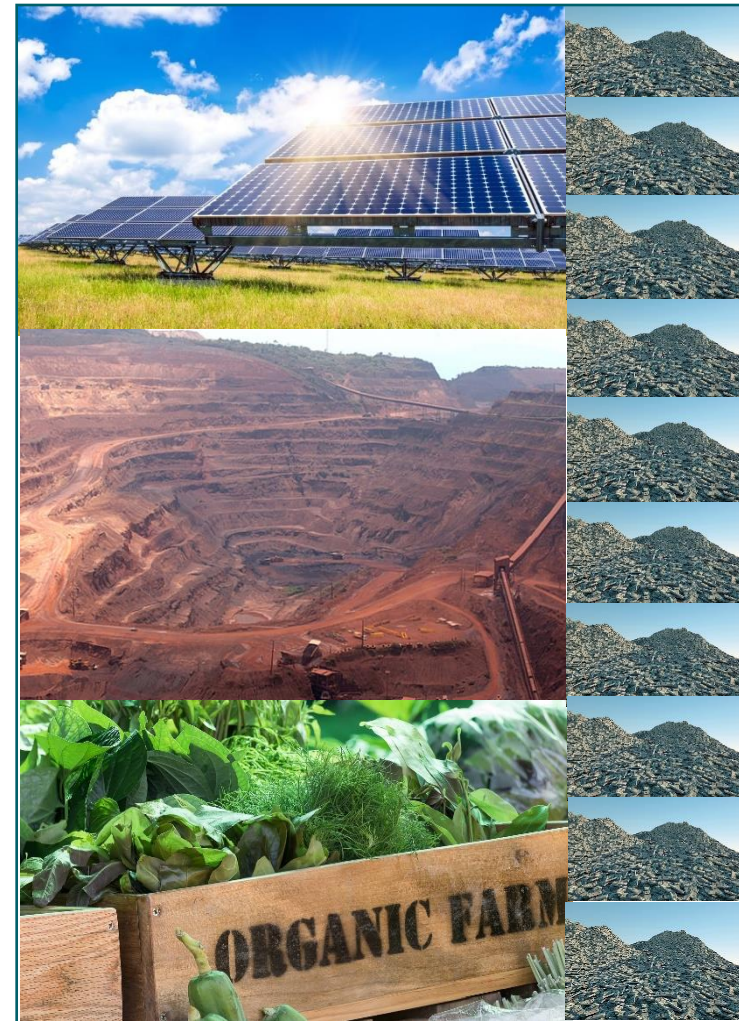
CBA of private projects



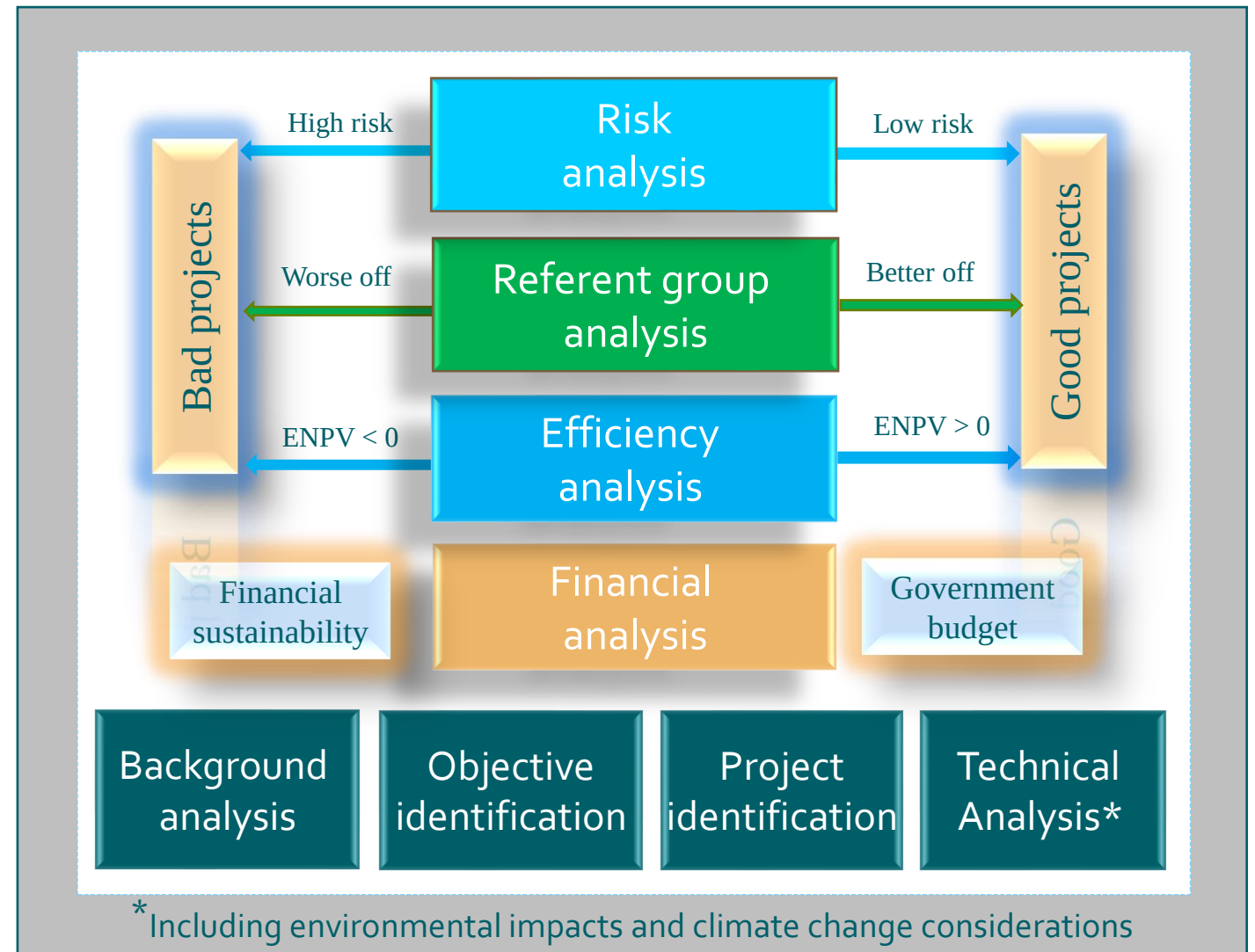
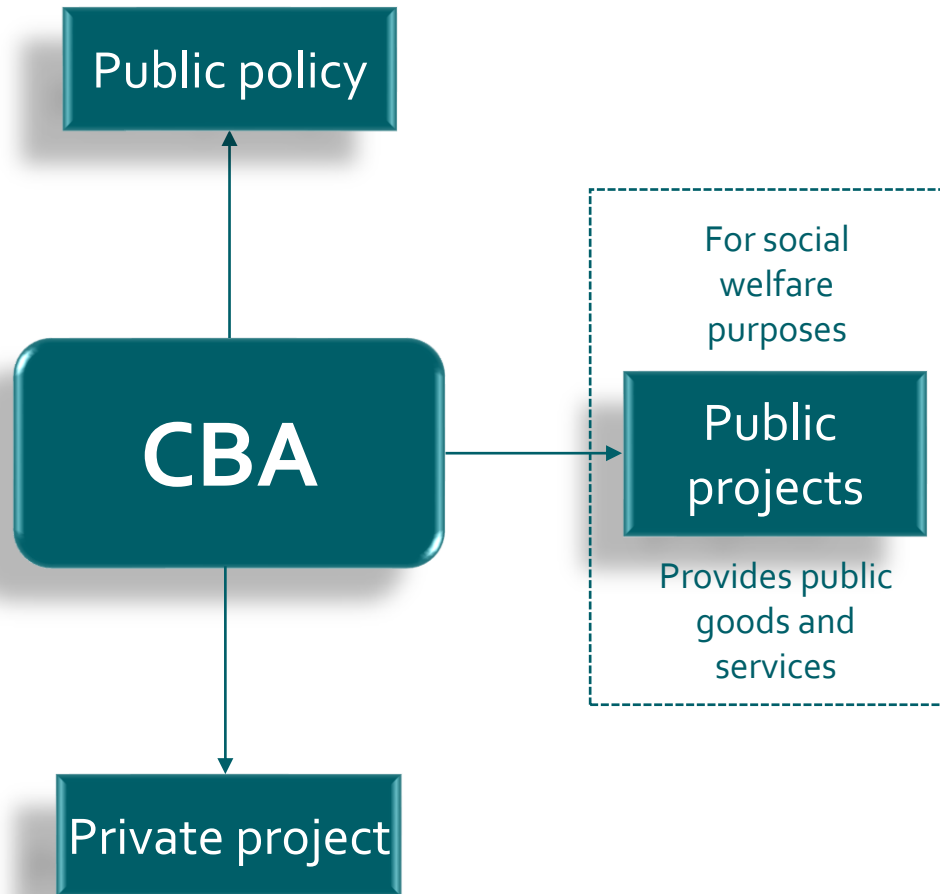
This is an integrated analysis, in which financial analysis plays a key role; while **efficiency** (or economic) and referent group analyses are only done if the private owners ask for concessions from the government. Efficiency analysis often uses standard conversion factors and/or adjusts distortions of market benefits and costs. This is called **private project appraisal**.

Cost-benefit analysis of private projects

- Private investors mainly pursue financial **profits** rather than community welfare.
- A private investment appraisal takes account only the benefits and costs of the project to the project's proponent – its effect on revenues and costs and hence on profit.
- Project costs such as land, labour, capital, materials, energy, ... are **cash outlays**, and project benefits are **cash receipts**. They are simply measured by **market prices**.
- If the project meets the local planning, environmental regulations, ... and performs a *minimum required rate of return* then the equity owners may invest in it.
- In cases of asking for concessions from the government, or creating negative externalities, efficiency and referent group analyses may be required.



CBA of public projects

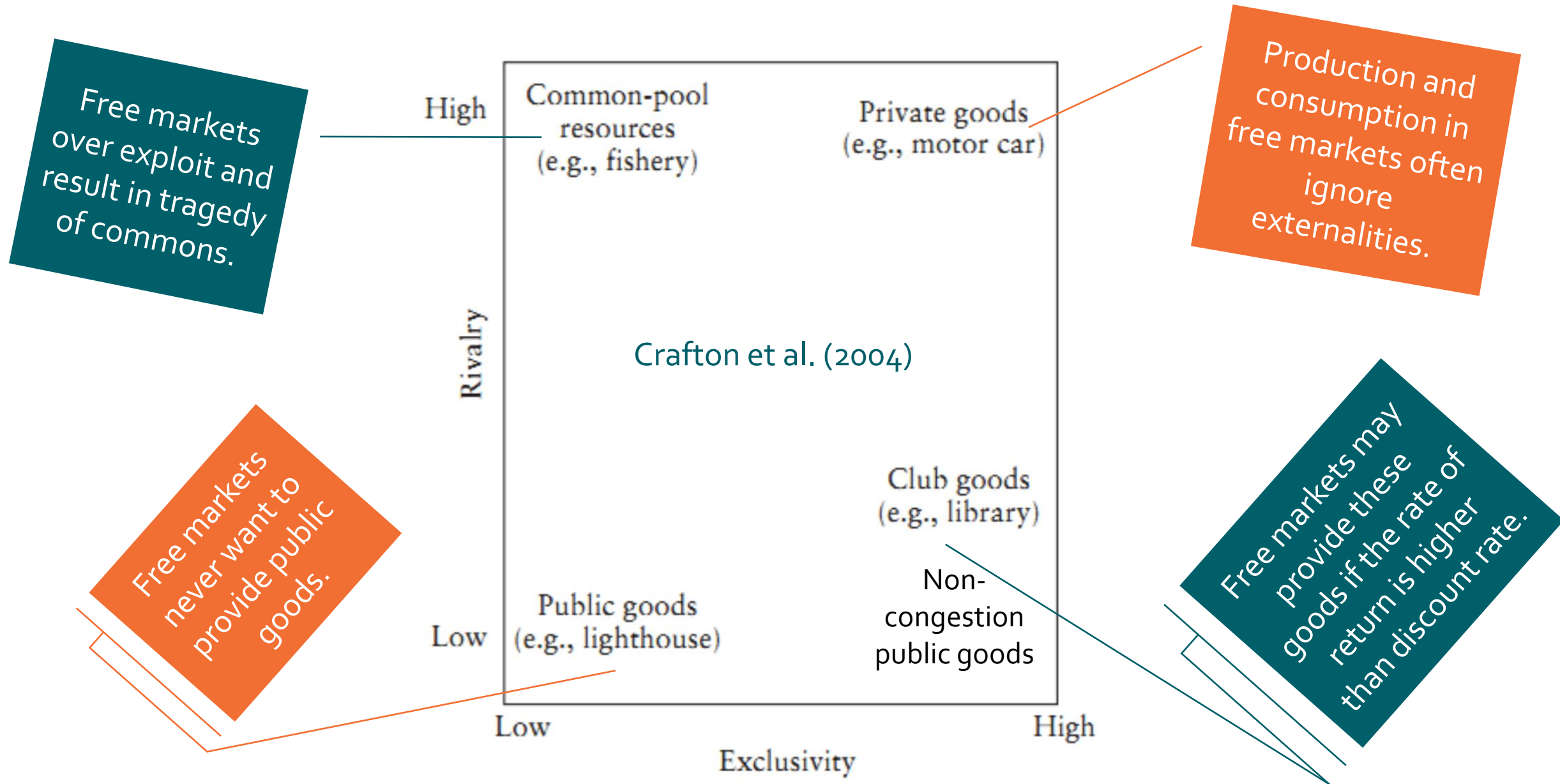


Due to budget constraints, governments are unable to implement all public projects that improve social welfare. Efficiency analysis provides a tool for making decisions from the society's perspective. In these cases, markets for project's outputs are often not available. Nonmarket valuation techniques may be used. This is called **public project appraisal**.

Cost-benefit analysis of public projects

- Governments directly provide public goods in order to improve **social welfare** under budget constraints.
- Public projects are often thought of in terms of the provision of **physical capital** in the form of infrastructure such as bridges, highways, dams, land reclamation, pollution control, fish stock enhancement and provision of parks.
- Other types of projects in forms of **human capital** (health, education, skill development), **social capital** (drug use prevention, crime prevention, the reduction of unemployment), and change in public policy (tax/subsidy, regulatory regime).
- A social CBA takes a wider perspective – it measures and compares the costs and benefits experienced by all groups affected by the project.





Introduction to project appraisal

Some basic concepts



Key concepts in project appraisal

Investment

Investment involves diverting scarce resources – land, labour, capital and materials – from the production of goods for current consumption to the production of capital goods which will contribute to increasing the flow of consumption goods available in the future.

Investment project

An investment project is a **particular allocation** of scarce resources in the present which will result in a flow of output in the future. For example, the construction of a dam which will result in increased hydroelectricity, irrigation, recreational opportunities and flood control in the future.

Key concepts in project appraisal

Costs

The cost of the project is measured as an **opportunity cost** – the value of goods and services would have been produced by the land, labour, capital and materials inputs had they not been used to undertake the project.

Benefits

The benefit of the project is measured as the value of the extra outputs produced by the project.

The monetary values of these costs and benefits depend remarkably on the appraisal points of view.

Key concepts in project appraisal

Appraisal

We use the word “appraise” in a **prospective sense**, referring to the progress of actually deciding whether resources are to be allocated to the project or not. In such a case, cost is the opportunity cost which will occur if the project is implemented.

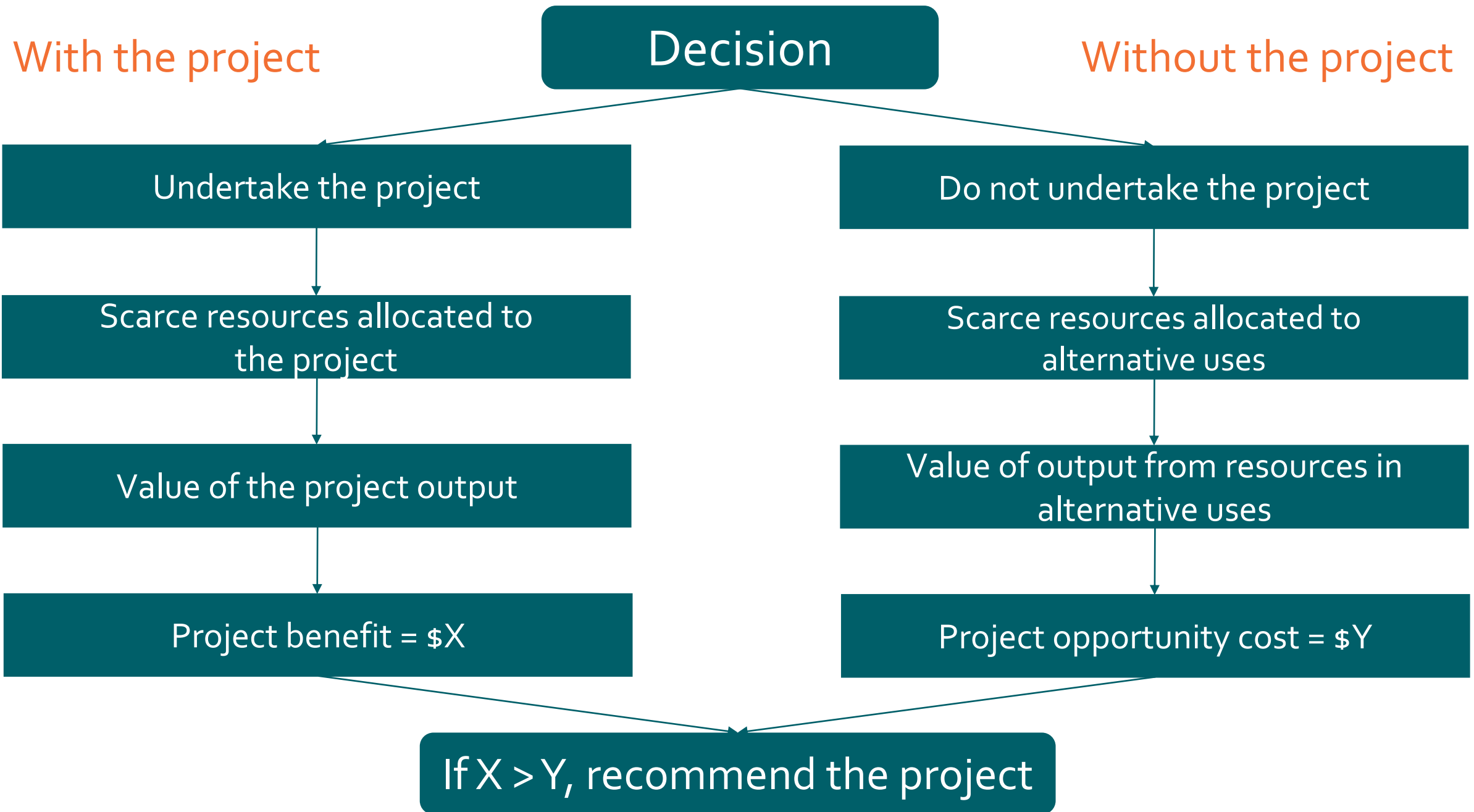
Evaluation

We use the word “evaluate” in a **retrospective sense**, referring to the progress of reviewing the performance of a project. In such a case, cost is the actual cost which already occurred because the project was really implemented.

Key concepts in project appraisal

With and without project

- An investment project makes a **difference** and the role of CBA is to measure that difference.
- There are two alternatives: undertake the project (with the project) and don't undertake the project (without the project).
- The ***with-and-without approach*** is at the heart of the CBA process and also underlies the important concept of opportunity cost.
- The world *without* the project is not the same as the world *before* the project; and the world *with* the project is not the same as the world *after* the project.

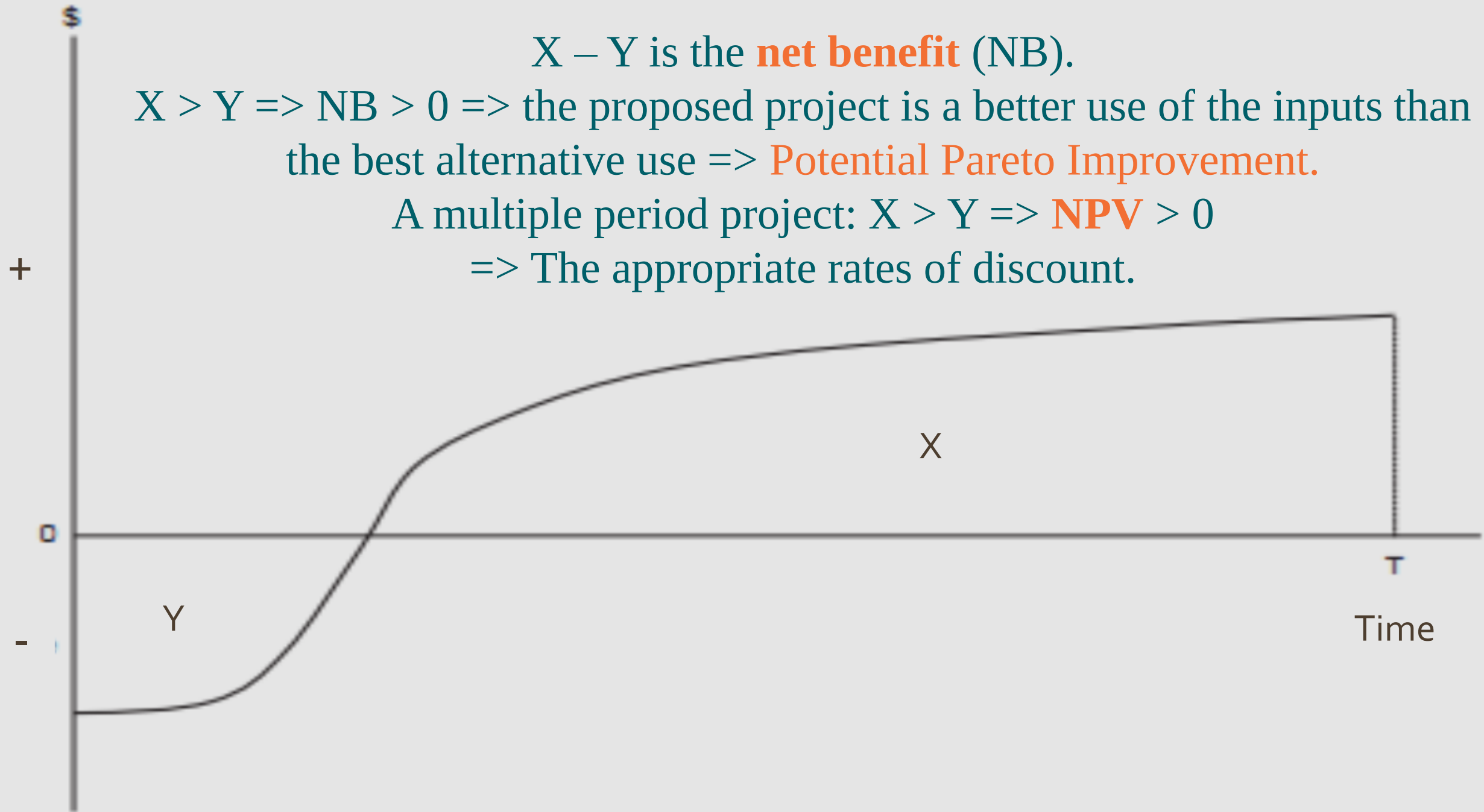


$X - Y$ is the **net benefit** (NB).

$X > Y \Rightarrow NB > 0 \Rightarrow$ the proposed project is a better use of the inputs than the best alternative use $\Rightarrow$ **Potential Pareto Improvement**.

A multiple period project: $X > Y \Rightarrow$ **NPV** > 0

$\Rightarrow$ The appropriate rates of discount.

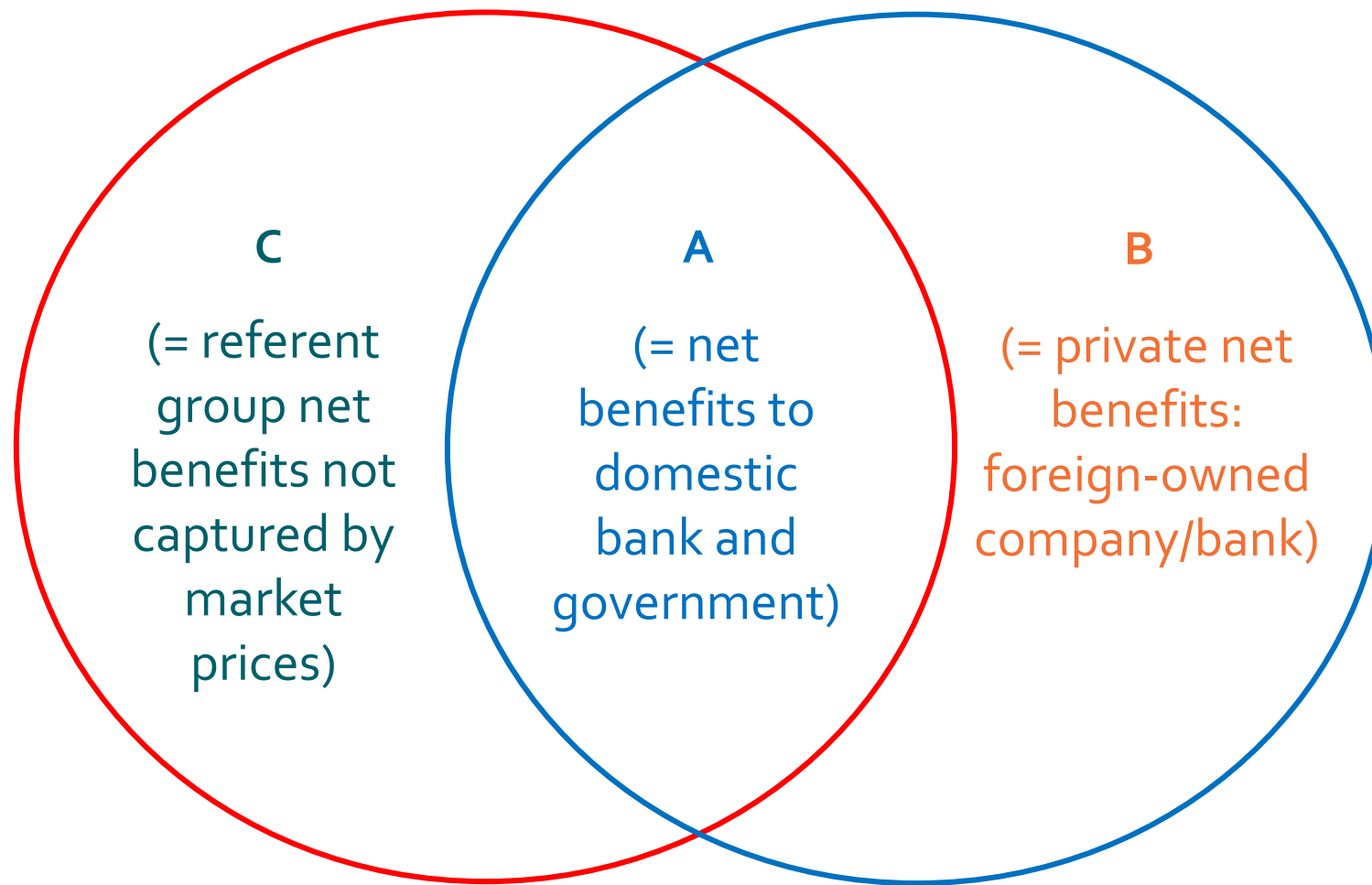


Introduction to project appraisal

Cost-benefit analysis model



Source: Campbell & Brown (2016)



A: Referent Group
(market prices)

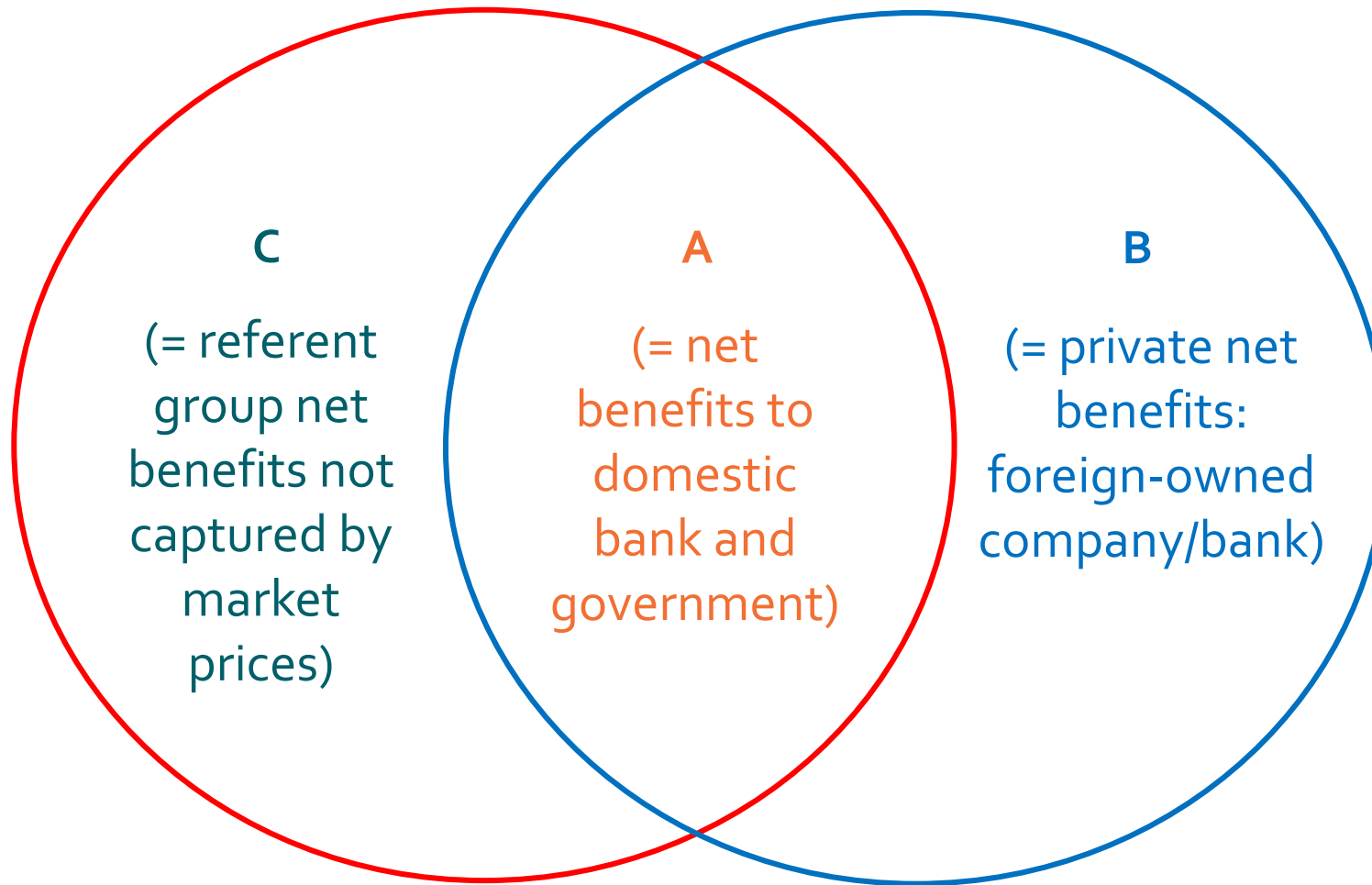
B: Non-referent
Group (market
prices)

C: Referent Group
(non-market
prices)

A+B: Market
(market prices)

A + B + C: Efficiency (market and non-market prices)

Source: Campbell & Brown (2016)



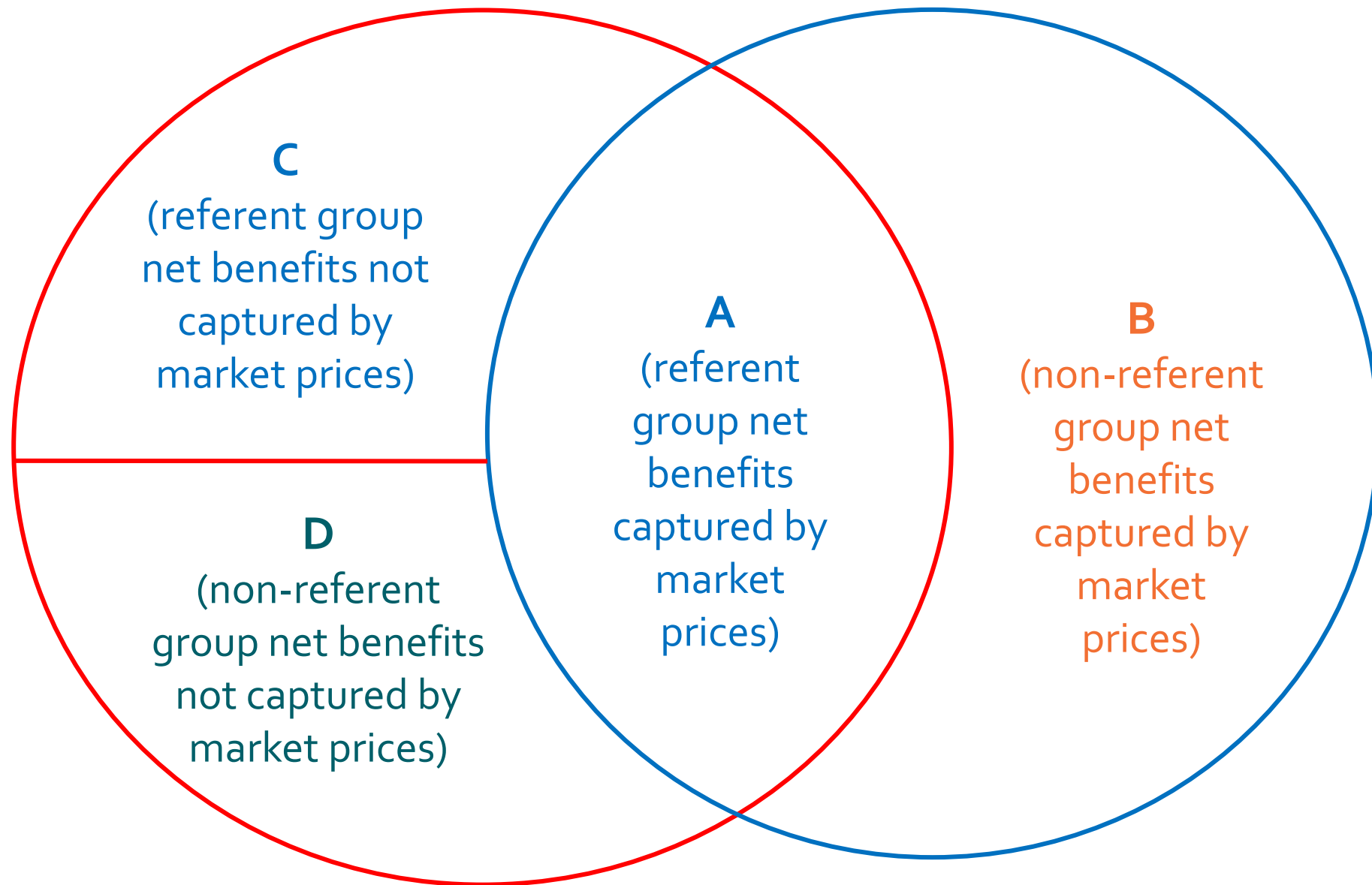
A: Referent Group
(market prices)

B: Private Analysis
(market prices)

C: Referent Group
(non-market
prices)

A+B:
Market/Project
Analysis (market
prices)

A + B + C: Efficiency/Economic Analysis (market and
non-market prices)



A + C: Referent Group Analysis

Source: Campbell & Brown (2016)

Introduction to project appraisal

The referent groups



The referent groups

- Before conducting the CBA study, the analyst needs to know whose benefits and costs are to be measured and compared.
- Not all the benefits and costs identified by the efficiency analysis are relevant to the decision about the project.
- The simplest way to determine the appropriate focus of the study is to ask the decision-maker to specify who is to be included and who is to be left out.
- Referent group identification is important in public projects, foreign-owned projects and large-scale projects of domestic investors.
- It is not necessary to identify referent group for stand-alone private projects which aim to evaluate the financial profitability.

The referent groups

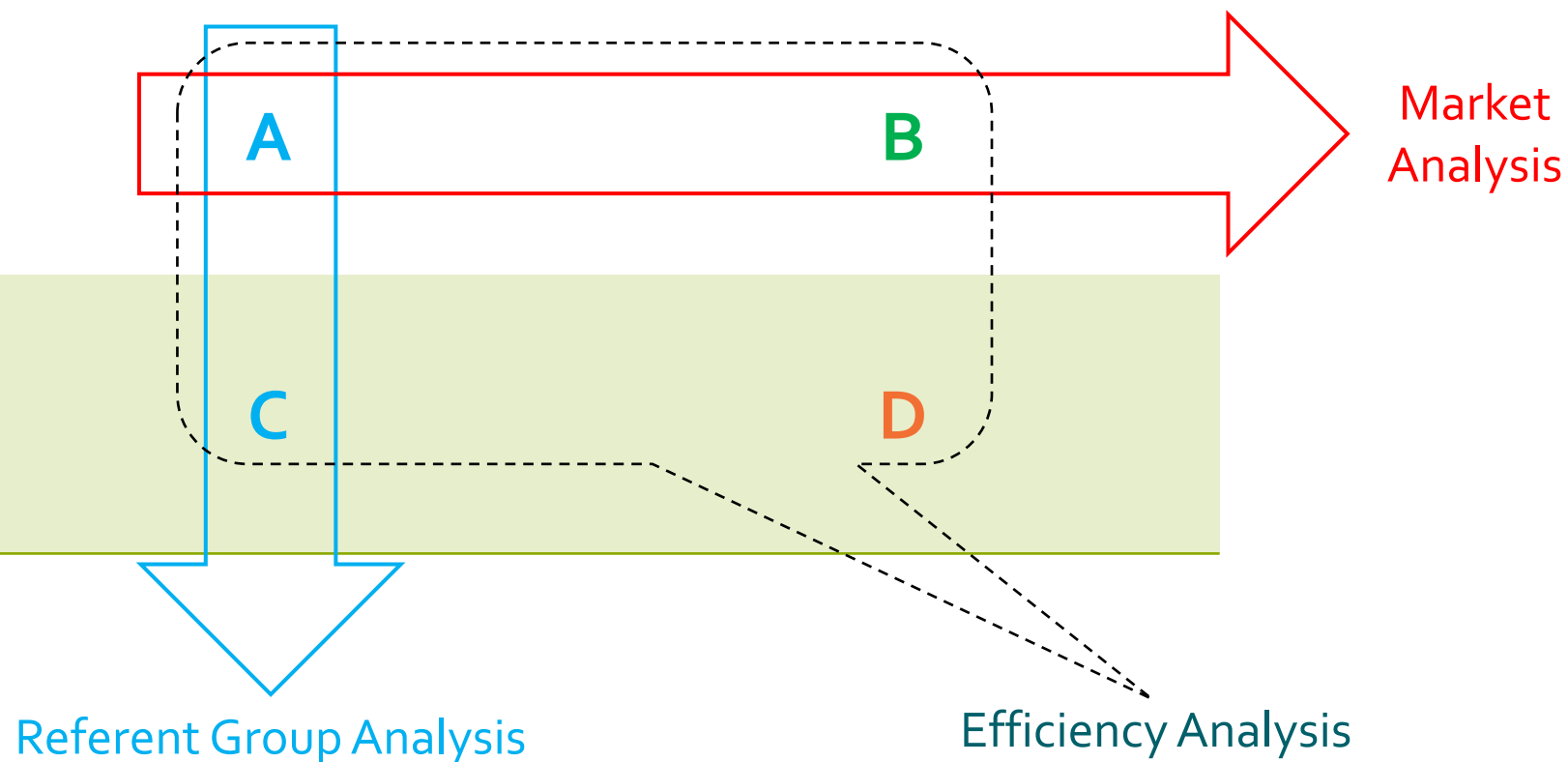
The project net benefits

Referent group

Non-referent group

Net benefits are measured by market prices

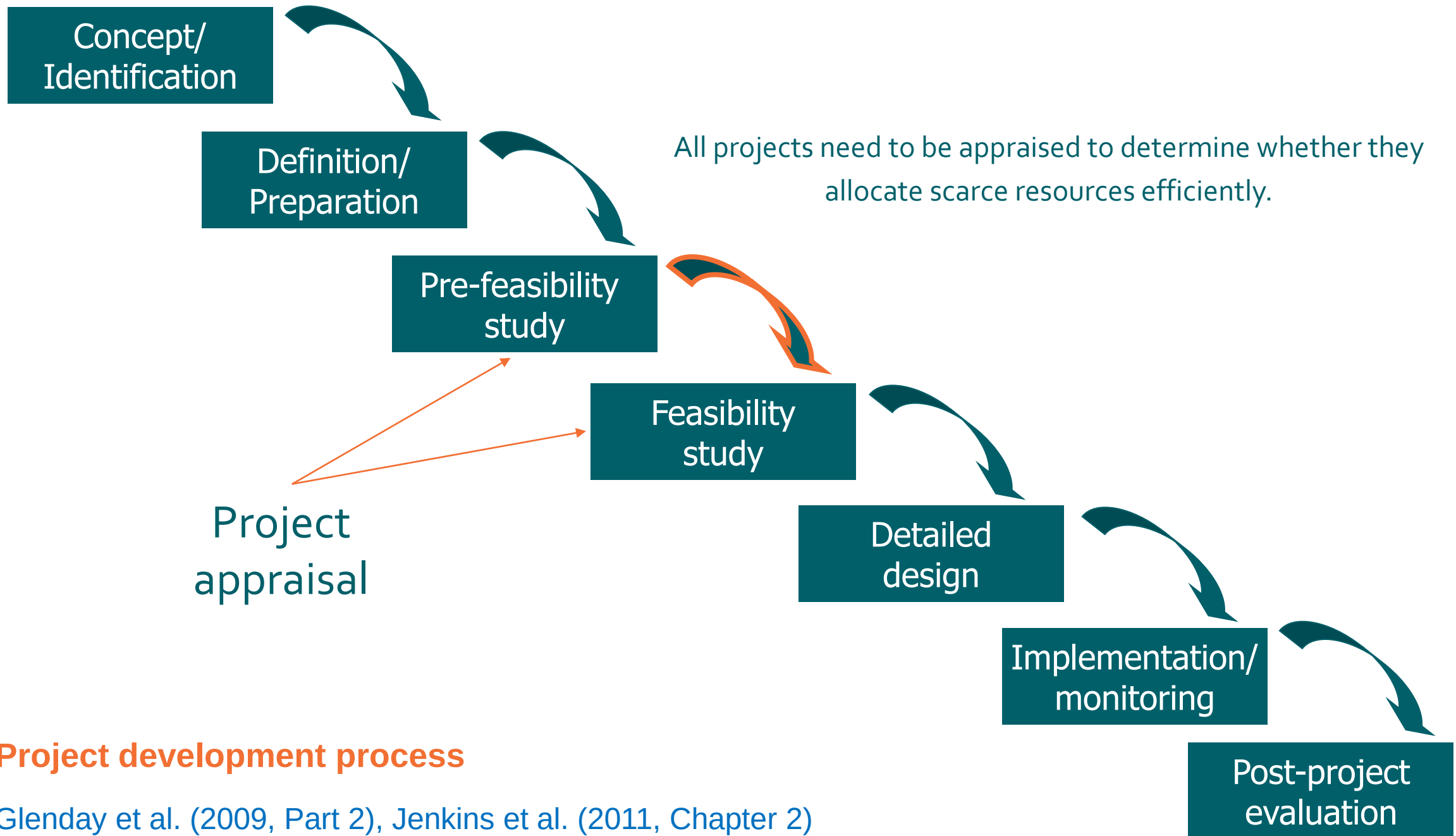
Net benefits are not measured by market prices



Introduction to project appraisal

Project development process





Project development process

Glenday et al. (2009, Part 2), Jenkins et al. (2011, Chapter 2)

Contents of a pre-/feasibility study

1. Demand/market module
2. Technical module
 - Environmental/climate change module
 - Human resource/organization module
 - Institutional/legal module
3. Financial module
4. Economic/efficiency module
5. Referent group/distributional module



Risk analysis
module

Contents of a pre-/feasibility study

Pre-feasibility study

- The first step in evaluating the overall attractiveness of the project. Its aim is to set up basics for feasibility study.
- Striking features:
 - Sustain the consistency of project data quality.
 - Use available secondary data.
 - For benefits, the analyst should use downward biased estimates; for costs, the analyst should use upward biased estimates.
 - Answer whether the project is feasible or not.

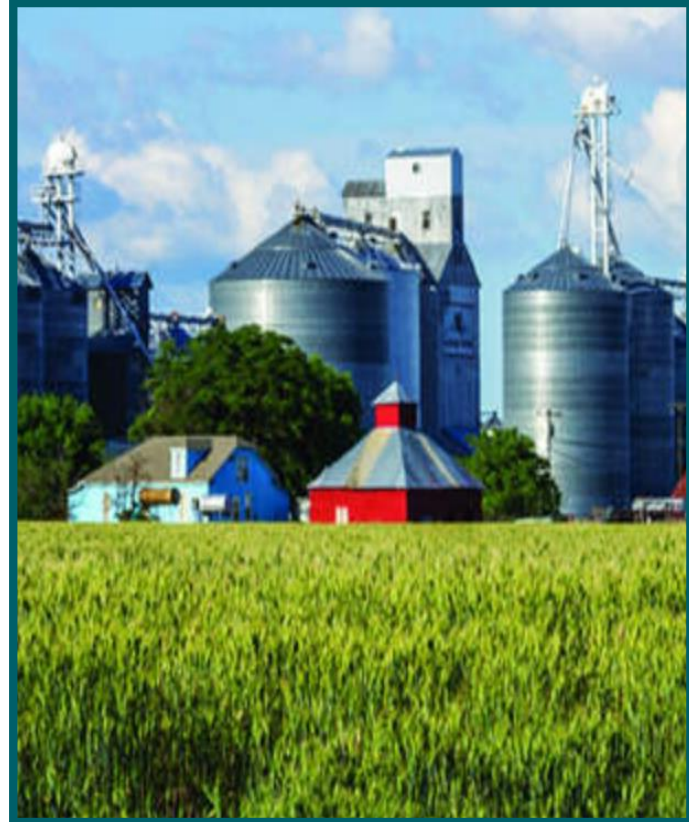
Feasibility study

- If the answer from the pre-feasibility study is feasible, the project may be studied in more detail.
- Striking features:
 - Improve the accuracy of key variables of the project.
 - Conduct surveys, collect primary data to re-calculate the background, market, technical, manpower, financial and economic analyses.
 - Analyze in detail the risk characteristics and recommend risk mitigation strategies.

After each study decision: accept, delay or reject the project proposal.

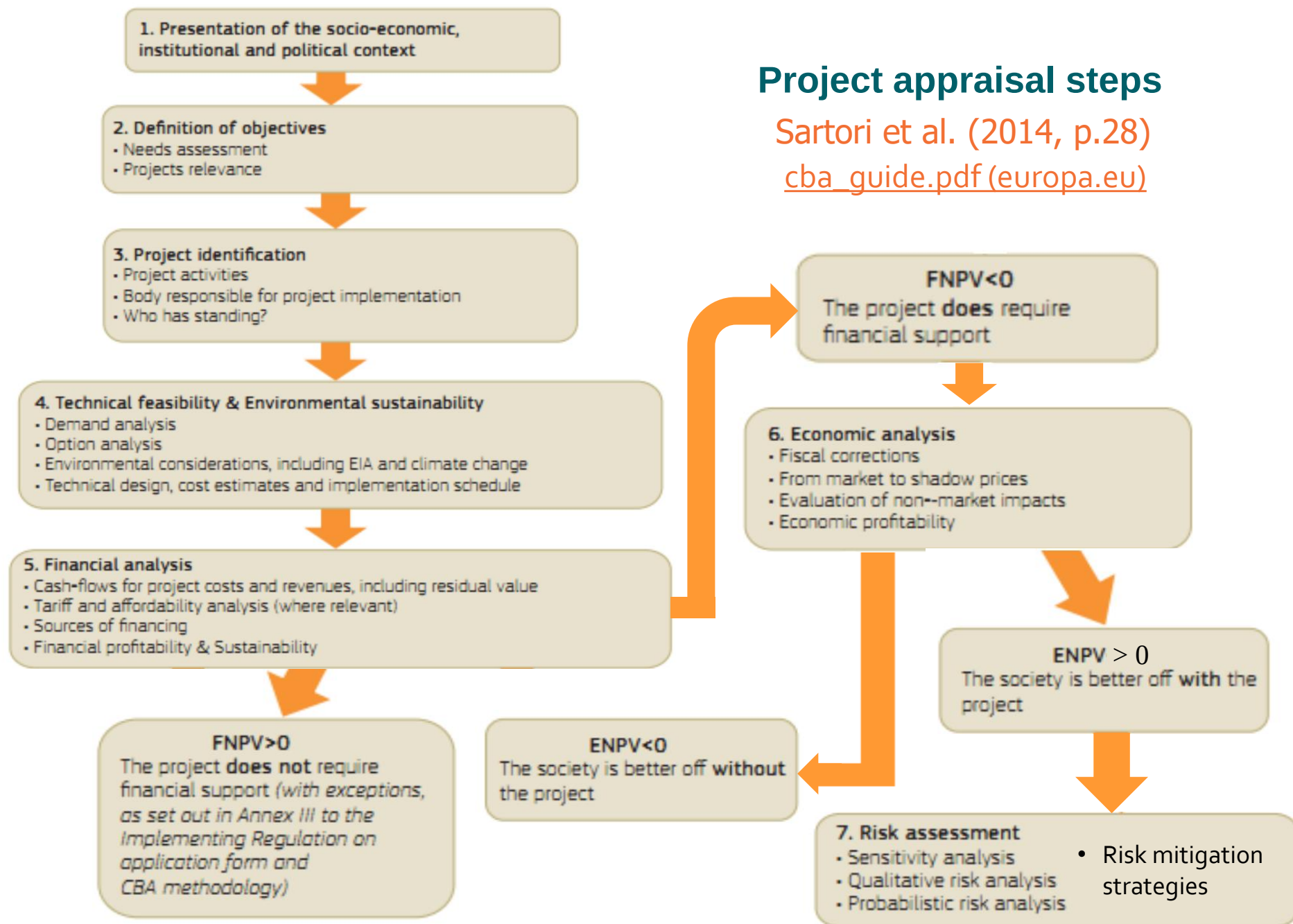
Introduction to project appraisal

Steps in a project appraisal



Steps in a project appraisal

- Clearly specify the project to be appraised:
 - Research the development plans of project-related industry or sector.
 - Determine location, time, referent groups; relate the project to other existing projects and programs, ...
 - Identify alternative options, including the status quo.
- List and describe quantitatively all impacts (i.e. inputs, outputs) of the project.
- Estimate in monetary values of benefits and costs (market, efficiency) of these impacts.
 - For market benefits and costs, the analyst estimates both market and efficiency prices by using economic theory.
 - For non-market benefits and costs (or zero market prices), the analyst may use non-market valuation techniques to estimate their economic values.
 - If the non-market values are hard to estimate due to budget constraint or other limitations, the analyst may use benefit transfer methods or ask experts for their opinions.
- Tabulate the benefits and costs (market, private, efficiency and referent group); calculate the alternative investment criteria such as NPV, IRR, BCR, ... and then compare among options.
- Perform risk analysis and offer recommendations.



Steps in a project appraisal

According to Campbell & Brown (2016, pp.9-10), a standard project appraisal process (both public & large-scale projects) for each option is as follows:

- 1) Project data
- 2) Market analysis (market prices)
- 3) Private analysis (market prices)
- 4) Efficiency analysis (i.e. social CBA, efficiency prices)
- 5) Referent group analysis (efficiency prices, both aggregate net benefit and disaggregated net benefits).

Steps in a project appraisal



Introduction to project appraisal

Three basic analyses



Types of cost-benefit analysis

-
- The diagram illustrates the types of cost-benefit analysis, categorized into two main groups: Financial and Economic. The Financial group includes Market CBA/Analysis and Private CBA/Analysis. The Economic group includes Efficiency CBA/Analysis and Referent Group CBA/Analysis. Each category is represented by a bracket on the left side of the slide, with the category name written vertically next to it.
- Financial**
 - **Market CBA/Analysis:** Market prices, financial discount rates, no externality, the project's total profitability from the standpoint of the total capital.
 - **Private CBA/Analysis:** Market prices, financial discount rates, no externality, the project's profitability from the standpoint of the owners' equity capital.
 - Economic**
 - **Efficiency CBA/Analysis:** Efficiency prices, social discount rates, externality, the overall economic efficiency of the project from the standpoint of the society/country.
 - **Referent Group CBA/Analysis:** Efficiency prices, social discount rates, externality, the distribution of project's net benefits to domestic citizens of a particular region/country.

Types of cost-benefit analysis?

- The “cost-benefit analysis” is a general term. Its name clearly indicates an analysis that aims to compare costs and benefits of a project.
- **Questions:** Who will burden costs of the project? Who will gain benefits of the project?
- The answer of these questions depends on how we look at costs and benefits; and distinguish financial analysis and economic/efficiency analysis.

Financial analysis of a project

- Financial analysis of projects is a part of CBA, where:
 - Takeholders in the analysis are the project's owners, sponsors (and government).
 - Costs in the analysis are financial costs (i.e. cash outlays).
 - Benefits in the analysis are financial benefits (i.e. cash receipts).
 - Just take into account market costs and benefits to the project's proponent.
- Financial analysis considers the profit generating capacity of the project for the investor and/or debt service coverage for the sponsor on the basis of cash flow analysis.
- **Questions:** Whether the project makes its owner better off? And/or whether the project itself repays the debt service?

Economic analysis of a project

- Economic/efficiency analysis of projects is a part of CBA, where:
 - Takeholders are all citizens of a particular region (even a country), not just investor and/or sponsor.
 - Costs in the analysis include all costs to the society as a whole (beyond those have market prices).
 - Benefits in the analysis include all benefits to the society as a whole (beyond those have market prices).
- **Questions:** Whether the project increases social welfare? Who will gain the benefits and who will bear the costs?

Types of cost-benefit analysis

	Financial analysis	Economic analysis
Viewpoint	Financial beneficiaries	Society as a whole
Benefits	Cash receipts (market prices)	WTP of consumers/users for the project's outputs (efficiency prices)
Costs	Cash expenditures (market prices)	Opportunity costs of scarce resources used by the project and economic values of negative externalities (efficiency prices)

Net benefits/Net present value		Economic analysis	
		+	-
Financial analysis	+	Accept	?
	-	?	Reject

Types of cost-benefit analysis

Financial analysis

- Mostly private projects and public projects in form of PPP
- Focus on **financial efficiency** of private sector (e.g. maximum financial NPV)
- Market costs and benefits to the private owner only
- Use market prices
- Exclude external effects
- Provide information for private decision makers.

Economic analysis

- Some private projects, all public projects and policy
- Focus on **economic efficiency** of the whole society (e.g. maximum economic NPV)
- Both market and non-market costs and benefits to everyone
- Use efficiency/shadow prices
- Include external effects
- Provide information for political decision makers and/or the public.

Introduction to project appraisal

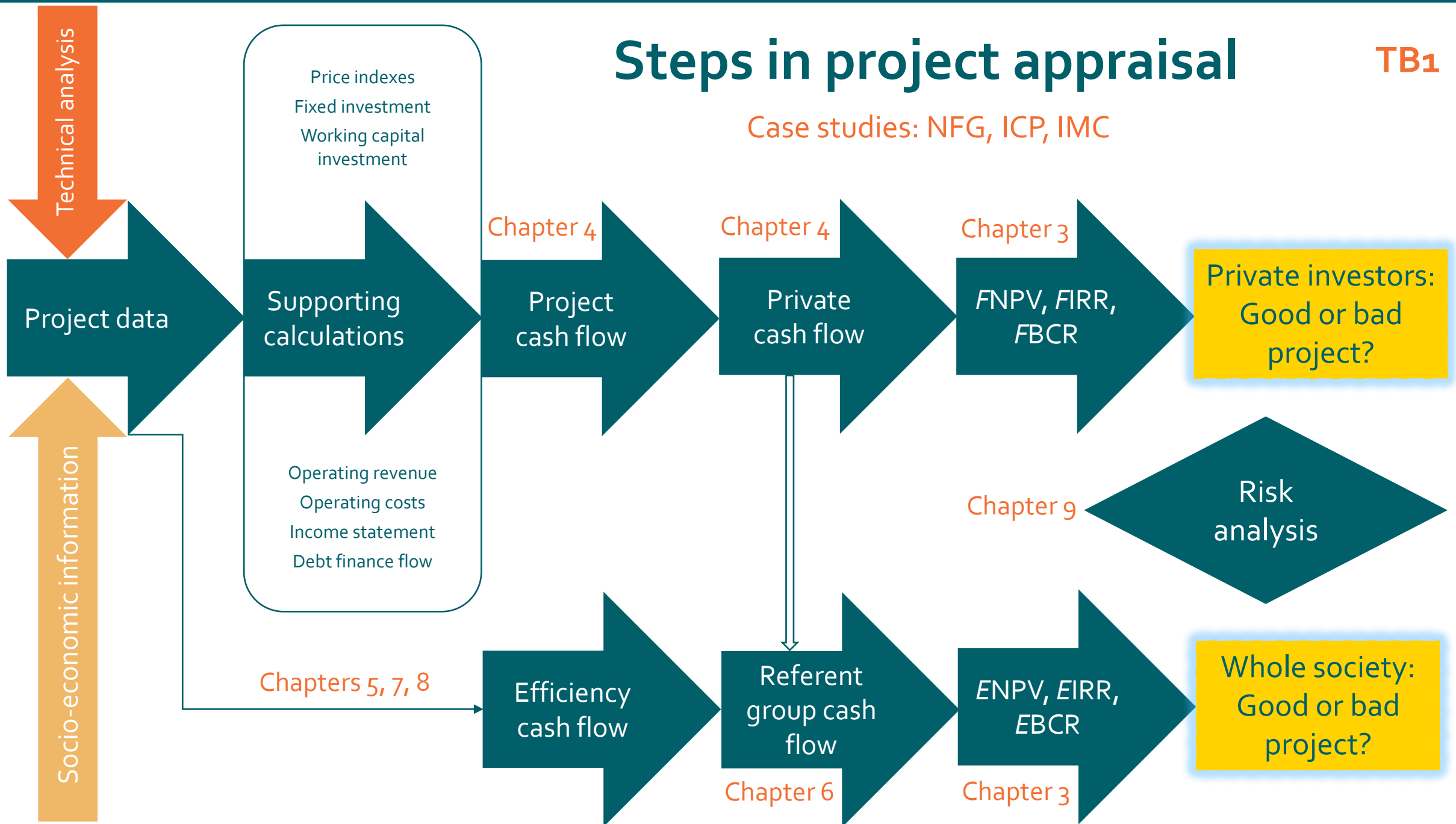
Using spreadsheets



Steps in project appraisal

TB1

Case studies: NFG, ICP, IMC



Introduction to project appraisal

An illustrative example



Section 1: Project Data						
Capital cost (market price)	100	Labour Input/year	20	Domestic loan	50	5 years
Capital cost (efficiency price)	93	Market wage	3	Interest rate	5%	/year
Output flow/year	10	Shadow wage	2	Pollution	25	/year
Product price (market)	10	Other costs	10	Tax rate	20%	/year
Product price (efficiency)	11	<i>(Market price = Efficiency price)</i>		Discount rates	5%	10%

Section 2A: Market Cost-Benefit Analysis (before taxes)

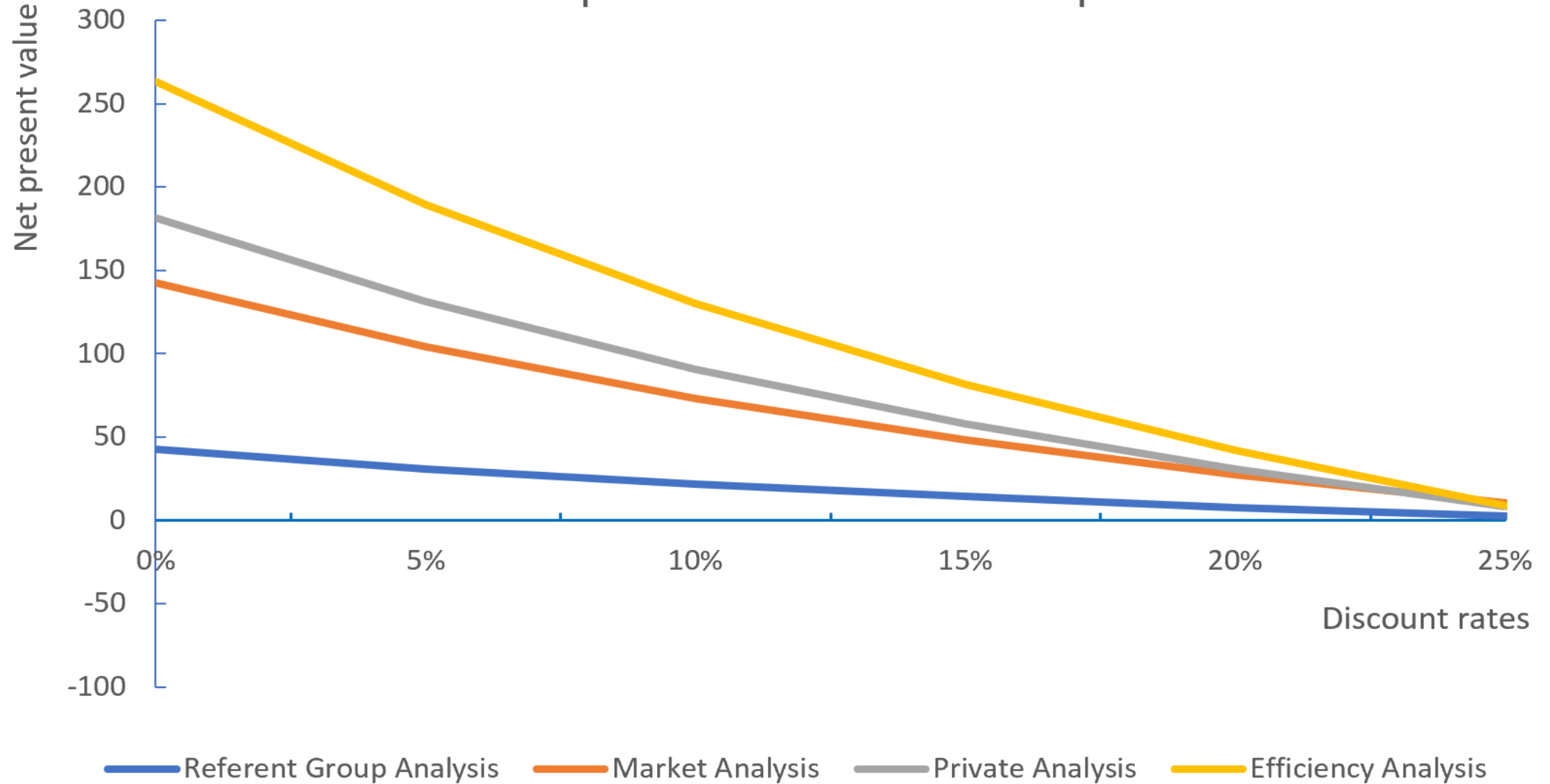
Year	0	1	2	3	4	5
Gross revenues		110	110	110	110	110
Capital costs	100					
Labour costs		60	60	60	60	60
Other operating costs		10	10	10	10	10
Net cash flow	-100	40	40	40	40	40.0
NPV@5% = 73.18		NPV@10% = 51.63		IRR = 29%		

Income tax						
Year	0	1	2	3	4	5
Net revenues		100	100	100	100	100
Labour costs		60	60	60	60	60
Other operating costs		10	10	10	10	10
Depreciation		25	25	25	25	25
Interest payments		2.5	2.0	1.6	1.1	0.5
Taxable income		2.5	3.0	3.4	3.9	4.5
Income tax		0.5	0.6	0.7	0.8	0.9

Section 2B: Market Cost-Benefit Analysis (after taxes)						
Year	0	1	2	3	4	5
Gross revenues		110	110	110	110	110
Capital costs	100					
Labour costs		60	60	60	60	60
Other operating costs		10	10	10	10	10
Sales tax		10	10	10	10	10
Income tax		0.5	0.6	0.7	0.8	0.9
Net cash flow	-100	30	29	29	29	29.1
	NPV@5% = 26.94	NPV@10% = 11.18		IRR = 14%		
Section 3: Private Cost-Benefit Analysis						
Year	0	1	2	3	4	5
Gross revenues		110	110	110	110	110
Loan	50					
Capital costs	100					
Labour costs		60	60	60	60	60
Other operating costs		10	10	10	10	10
Loan repayments		11.5	11.5	11.5	11.5	11.5
Sales tax		10	10	10	10	10
Income tax		0.5	0.6	0.7	0.8	0.9
Net cash flow	-50	18	18	18	18	17.6
	NPV@5% = 26.94	NPV@10% = 17.40		IRR = 23%		

Section 4: Efficiency Cost-Benefit Analysis						
Year	0	1	2	3	4	5
Benefits		110	110	110	110	110
Capital costs	93					
Labour costs		40	40	40	40	40
Other operating costs		10	10	10	10	10
Pollution		25	25	25	25	25
Net benefit flow	-93	35	35	35	35	35
NPV@5% = 58.07		NPV@10% = 39.22		IRR = 25%		
Section 5: Referent Group Cost-Benefit Analysis (e.g. equity owner is a foreign company)						
Year	0	1	2	3	4	5
RG aggregate net benefit flow	-43.5	17.0	17.1	17.2	17.3	17.4
RG net benefit allocation						
Domestic bank	-50.0	11.5	11.5	11.5	11.5	11.5
Government	6.5	10.5	10.6	10.7	10.8	10.9
Import duty	6.5					
Sales taxes		10.0	10.0	10.0	10.0	10.0
Income taxes		0.5	0.6	0.7	0.8	0.9
Labour (rent)		20.0	20.0	20.0	20.0	20.0
Community (pollution)		-25.0	-25.0	-25.0	-25.0	-25.0
Total RG net benefit flow	-43.5	17.0	17.1	17.2	17.3	17.4
Check for consistency	0.0	0.0	0.0	0.0	0.0	0.0
NPV	0%	5%	10%	15%	20%	25%
Referent Group Analysis	42.7	31.1	21.8	14.2	8.0	2.8
Market Analysis	100.0	73.2	51.6	34.1	19.6	7.6
Private Analysis	38.8	26.9	17.4	9.6	3.2	-2.1
Efficiency Analysis	81.5	58.1	39.2	23.9	11.2	0.7
Check for consistency	42.7	31.1	21.8	14.2	8.0	2.8

A comparison of different viewpoints



Project appraisal

Q&A section

